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24. def main():
25.     """Run a single game of Hacking."""
26.     print('Hacking Minigame, by Al Sweigart al@inventwithpython.com
27. Find the password in the computer's memory. You are given clues after
28. each guess. For example, if the secret password is MONITOR but the
29. player guessed CONTAIN, they are given the hint that 2 out of 7 letters
30. were correct, because both MONITOR and CONTAIN have the letter O and N
31. as their 2nd and 3rd letter. You get four guesses.\n''')
32.     input('Press Enter to begin...')
33.
34.     gameWords = getWords()
35.     # The "computer memory" is just cosmetic, but it looks cool:
36.     computerMemory = getComputerMemoryString(gameWords)
37.     secretPassword = random.choice(gameWords)
38.
39.     print(computerMemory)
40.     # Start at 4 tries remaining, going down:
41.     for triesRemaining in range(4, 0, -1):
42.         playerMove = askForPlayerGuess(gameWords, triesRemaining)
43.         if playerMove == secretPassword:
44.             print('A C C E S S   G R A N T E D')
45.             return
46.         else:
47.             numMatches = numMatchingLetters(secretPassword, playerMove)
48.             print('Access Denied ({} / 7 correct)'.format(numMatches))
49.     print('Out of tries. Secret password was {}'.format(secretPassword))
50.
51.
52. def getWords():
53.     """Return a list of 12 words that could possibly be the password.
54.
55.     The secret password will be the first word in the list.
56.     To make the game fair, we try to ensure that there are words with
57.     a range of matching numbers of letters as the secret word."""
58.     secretPassword = random.choice(WORDS)
59.     words = [secretPassword]
60.
61.     # Find two more words; these have zero matching letters.
62.     # We use "< 3" because the secret password is already in words.
63.     while len(words) < 3:
64.         randomWord = getOneWordExcept(words)
65.         if numMatchingLetters(secretPassword, randomWord) == 0:
66.             words.append(randomWord)
67.
68.     # Find two words that have 3 matching letters (but give up at 500
69.     # tries if not enough can be found).
70.     for i in range(500):
71.         if len(words) == 5:
72.             break # Found 5 words, so break out of the loop.
73.
74.     randomWord = getOneWordExcept(words)
75.     if numMatchingLetters(secretPassword, randomWord) == 3:
76.         words.append(randomWord)
77.

```